Engineering Mathematics - I · Unit_01 · Unit 1

DRAFT — UE26MA141A · Unit_01 · Unit 1 · Problems on Maxima and Minima for a function of two variables

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Problems on Maxima and Minima for a function of two variables

1 Problems on Maxima and Minima for a function of two variables

Study focus: Problems on Maxima and Minima for a function of two variables. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000222 · EV0000223

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we solve problems on maxima and minima for a function of two variables.

Key details

- F of x comma y.
- Find the maximum and minimum values of function x power 3 plus 3xy square minus 15x square minus 15y square.
- As per the procedure, we first find out the partial derivative of f with respect to x and y, which we call them as p as well as q.
- Next, in order to find out critical points, we need to equate fx as well as f y to zero.
- From f y equal to zero, we see that either y equal to zero or x equal to five.
- It means when we equate 6x minus 30y to 0, we see that either y 0 or x equal to 5.
- We have one more case left out.
- In a similar manner, we have to put x equal to 5 in expression number 1.
- Then we need to compute R S t.
- R is nothing but dou square f divided by dou x square, which is 6x minus 30.
- Then S is nothing but dou square f divided by dou x dou y which is nothing but 6y.
- So therefore R T minus s square becomes 36 of x minus 5 whole square minus pi square.
- As we see that there are four critical points.
- So we can think of four cases because of four critical points.
- At 0, 6 comma 0 we see that R t minus s square is positive, but R is positive.
- Therefore, we can say that at 6 comma 0 the function is minimum.

- At the critical point 4 comma zero, what happens?
- At 4 comma zero, R T minus S square is 36, which is positive, but R is negative.
- Therefore, we say that at the point 4 comma 0, the function is maximum.
- Therefore we say that we say that these are the two stattle points.

Example or application from the lecture

- Let's start with this problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000222 · EV0000223 · EV0000224 · +1 source units in pack.json

Core explanation · 05:00–09:59

We first found out PQ, then we equated P and Q to 0.

Key details

- Then we found out critical points.
- Then we computed RT minus S square.
- Then evaluated RT when C square at each of those critical points then arrived at this conclusion.
- Find the highest temperature on the surface of the sphere x square plus y square plus z square equal to a square.
- Note that because we are looking for maximum temperature, we assume that k to be a positive constant.
- We need to convert this function of three variables to a function of two variables by using z value.
- P of x y z becomes k x y of a square minus x square minus y square.
- The value of R is going to be minus 6K XY.
- The value of S is going to be K of A square minus 3x square minus 3y square.
- The value of T is going to be minus 6 Kxy.
- Therefore, we say that 0.0 is a saddle point.
- In a similar manner, we see that at the points a by 2, a by 2, as well as minus a by 2, and minus a by 2, R T.
- Minus S square is positive, but R is negative because we are assuming K to be positive.
- Test the function for maxima, minima and settle point.
- Fy is going to be 4y power 3 minus 2y.
- Therefore, we say that the function is maximum at the point zero comma zero and the maximum value is one.
- In a similar manner, it is easy to see that at all these four points, R T minus S square is positive, but R is also positive, therefore we say that F is minimum at these points.
- Lastly, at plus r minus 1 by root 2 0, and 0, plus r minus 1 by root 2, we can see that r t minus s square is minus 8, which is negative, but r is also negative.
- Therefore, we say that these are the two saddle points.
- F of x comma y to be sine x plus sine of y plus sine of x plus y.

Example or application from the lecture

- Let's look at another problem.
- The temperature capital T at any point XYZ in space is given by this expression KXY Z square where K is a constant.
- Let's evaluate R T minus S square at each of these critical points.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000226 · EV0000227 · EV0000228 · +1 source units in pack.json

Core explanation · 10:00–14:59

Fy is going to be cos y plus cos of x plus y.

Key details

- Subtracting 2 from 2, we see that x equals y after cancellation of plus of x plus y and plus of x plus y.
- Then substitute x as y in expression 1, we see that x equal to 5 by 3.
- So because x is y.
- the only critical point is pi by three comma pi by three so at pi by three comma pi by three we see that after evaluat r t minus s square is positive but r is negative therefore we say that.

Example or application from the lecture

- The function is maximum at the point pi by three, comma pi by three, the maximum value is given by three root three divided by two.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000230 · EV0000231 · EV0000232 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Problems on Maxima and Minima for a function of two variables transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__e276ddb59439e6ec.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Maxima and Minima for a function of two variables transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__e276ddb59439e6ec.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Maxima and Minima for a function of two variables transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__e276ddb59439e6ec.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9962; only the JSON segmentation supplies the displayed timestamps.
Problems on Maxima and Minima for a function of two variables transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__e276ddb59439e6ec.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.

3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00058	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0039. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00059	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0039. Sidecar variants compare...
SRC00060	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0039. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00061	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0039. Sidecar variants compare...

C Revision and retrieval practice